

Tinker 2026 Team Description Paper

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Abstract. This paper presents an overview of Tinker-FuRoC’s domestic service robot for the RoboCup@Home League, including its system architecture and key technologies. Building upon our prior research in articulated object manipulation and sim-to-real transfer, we integrate learning-based methods with tightly coupled perception, navigation, manipulation, and task planning modules to enable robust real-world deployment. The system supports 3D visual and semantic navigation, multi-modal human tracking and following, learning-based grasping, and behavior-tree- and LLM-driven decision making to accomplish tasks such as Receptionist, Storing Groceries, Help Me Carry, and GPSR. To promote reproducibility and community development, all major software components are released as open source.

1 Introduction

The Future Robotics Club (FuRoC) is a student-led organization at Tsinghua University, composed primarily of undergraduate students from various departments such as Electronic Engineering, Software Engineering, Mechanical Engineering, Automation, and Computer Science. As one of its founding teams, Tinker focuses on the development of domestic service robots with practical and scalable artificial intelligence capabilities.

Over the years, Team Tinker has built a comprehensive robotic platform integrating perception, manipulation, speech interaction, and autonomous navigation. These core capabilities are further enhanced through the integration of multi-modal large language models (MLLMs), fine-tuned vision models, semantic navigation, and structured behavior tree-based decision systems. Through continuous iteration driven by both research and competition, the system has evolved to support an increasing range of complex household tasks.

Tinker has maintained a long-term presence in RoboCup@Home, achieving 5th place in 2016, 7th place in 2019, and 8th place in 2025. The 2026 season will mark our 10th participation in the World RoboCup@Home League. We also actively contribute to the RoboCup community, with one of our team members

serving on the RoboCup@Home organizing committee. Motivated by both technical progress and academic exchange, we participate again to benchmark our system against international standards and to contribute to the advancement of domestic service robotics.

2 Scientific Research

This section highlights the main research contributions of our team in domestic robotics, with a focus on learning-based manipulation and simulation-to-reality transfer.

2.1 Enhancing Articulated Object Manipulation

Recent advancements in domestic robotics have significantly enhanced the manipulation of articulated objects. We have developed two innovative frameworks which leverage the strengths of learning methodologies and sim-to-real approaches.

Sim2Real²[1] introduces an innovative method for manipulating unseen articulated objects in real scenarios without human guidance. It builds an interactive physics model for long-horizon manipulation trajectory planning using only two RGBD images. Experimental results show a high success rate in manipulating articulated objects, with less than 30% error, and the ability of advanced manipulation including tool use.

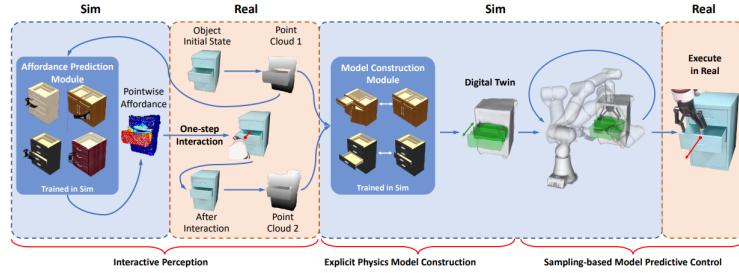


Fig. 1: Sim2Real²

Parallel to this, our Part-Guided 3D RL framework enhances the manipulation of unseen articulated objects using visual input [2]. It merges 2D segmentation with 3D reinforcement learning, improving RL policy training efficiency. A novel Frame-consistent Uncertainty-aware Sampling strategy enables the training of a single RL policy for multiple tasks, showing strong generalizability in simulated and real-world environments.

Together, these two frameworks mark a significant advancement in the field of domestic robotics. They showcase the potential of integrating learning techniques and sim-to-real approaches to improve the precision and adaptability of robots.

2.2 Revolutionizing Manipulation Skills with ManiSkill3

To support large-scale learning of general manipulation skills for service robots, we contributed to the development ManiSkill3 [3], a GPU-parallelized simulation platform for visual robotic manipulation.

ManiSkill3 significantly expands task diversity, environment complexity, and simulation realism, covering stationary and mobile manipulation, single- and dual-arm coordination, and long-horizon interactions with both rigid and articulated objects. A re-engineered simulation and rendering pipeline enables large-scale parallel data generation, supporting high-throughput reinforcement and imitation learning with realistic perception.

Our work provides a scalable foundation for training and evaluating manipulation policies intended for real domestic robots, propelling learning in robotic manipulation within the realm of home service robots, and setting new standards for simulation platforms.

3 Technical Contributions

In this section, we present the main technical components of the Tinker system and how we integrate existing methods with our own developments to achieve robust performance in household scenarios. Our focus is on improving task success rates and robustness under real-world disturbances. We also release key components as open source, including decision-planning repositories, a YOLO-based on-site segmentation pipeline, human-tracking with ReID, and ROS2-based simulation assets.

3.1 3D Visual and Semantic Navigation

Our navigation stack is built on Nav2 and extended with behavior trees to orchestrate multiple modular servers for high-level navigation logic. In the local planner, we employ the Spatial-Temporal Voxel Layer (STVL) as a 3D perception plugin in the local costmap, which improves modeling of indoor environments with complex geometry.

Tinker is equipped with a compact 3D LiDAR providing 360-degree perception. We apply a slicing scheme along the vertical axis to segment the point cloud into height bands and fuse them for obstacle detection. This yields true spatial obstacle avoidance, improving safety over traditional 2D approaches, especially around irregularly-shaped obstacles.

For localization, we use an RGB-D-based visual SLAM pipeline built on RTAB-Map [4]. This module constructs a dense, colored 3D point cloud map that serves as the basis for semantic reasoning. On top of this representation, we are integrating language-driven semantic navigation that associates map regions with language prompts, aiming to support more natural human-robot interaction in domestic environments.

3.2 Human Tracking

We employ a unified RGB-D human tracking framework that combines YOLO-based instance segmentation, short-term association, and a custom appearance-based ReID stack for robust identity preservation in crowded environments.

The front end uses a YOLO11-seg detector with ByteTrack [5] for short-term association. For long-term identity consistency, we integrate a ReID module with a ResNet50 backbone enhanced by channel attention and part-based pooling. In addition to deep embeddings, multi-scale color histograms in LAB/HSV space and upper/lower-body clothing descriptors are incorporated. All cues are fused by a weighted similarity aggregator with hard margins to suppress ambiguous matches.

Anchor embeddings are refreshed only under high-confidence matches to maintain stable identity prototypes. Confirmation windows and pre-confirmation streaks prevent rapid ID flipping. Camera-motion detection gates spatial heuristics, and occlusion handling caches the last reliable appearance for strict re-matching upon re-emergence.

Depth from the RGB-D stream is fused with segmentation masks to compute accurate 3D centroids at 15–20 Hz. During initialization, several reference frames are captured to establish ReID embeddings. If the target is lost for more than 10 s, the robot issues an audio prompt for the user to stop, during which it performs a rotational scan and re-identifies the user via ReID similarity. This multi-modal pipeline provides robust competition-grade human tracking for RoboCup@Home. Implementation is available as part of our 2026 vision module: `tk26_vision`

3.3 Task Planning via Behavior Trees

Designing decision systems for complex robotic tasks is challenging due to frequent module updates and evolving APIs. To manage this complexity, we adopt a behavior-tree-based decision architecture and open-source all components of our decision system. Our latest implementation was deployed at the Zhongguanchun Embodied AI Competition, where Team Tinker achieved 1st place in the service track.

Using `py_trees_ros`, we encapsulate primitive actions in modular nodes and dynamically construct behavior trees in response to natural-language commands. This enables flexible yet interpretable task planning and supports complex tasks such as the General Purpose Service Robot (GPSR) challenge. The system integrates robust low-level modules (navigation, perception, manipulation) with LLM-based reflective planning for reliability and adaptability. All related code is hosted under our GitHub organization, `TinkerFuroc`.

3.4 Vision Training Pipeline

Many RoboCup@Home tasks and domestic-service scenarios require fast adaptation to new household objects. To support on-site dataset creation and labeling,

we developed a high-throughput vision training pipeline that combines GroundingDINO’s [6] open-set vocabulary recognition with SAM2’s [7] segmentation-tracking capabilities, along with human-friendly error correction and re-masking features. Using a portable RealSense d435 camera, the system can collect segmentation masks at 60 FPS, producing training-ready data streams in real time.

This pipeline enabled us to train a YOLOv11-Seg (Medium) model on more than two dozen objects on site in under two hours, adapting to competition-specific item sets. With the release of Meta’s SAM3 [8], we are updating the pipeline to leverage its improved robustness. The full implementation is available at: `tk25_vision_train`.

3.5 Open-Source Simulation

To accelerate the development and testing of robust behaviors, we built an open-source simulation framework based on NVIDIA Isaac Sim. Our goal is to maintain a high-fidelity digital twin of the Tinker robot for large-scale data collection and algorithm validation in virtual environments. This reduces dependency on physical hardware and allows safe testing of edge cases that are difficult or risky to reproduce on the real system.

The simulation environment models lighting changes, sensor noise, and dynamic obstacles. We use it to generate training datasets for perception modules and to evaluate navigation and manipulation policies under diverse conditions. The simulator operates synchronously with ROS2, facilitating direct transfer of validated algorithms to the physical robot. By open-sourcing our simulation assets and pipelines, we aim to support reproducible research and community development.

3.6 Grasping

We developed a complete grasping system enabling Tinker to reliably manipulate a wide range of everyday objects. For grasp pose estimation, we employ both Contact-GraspNet and AnyGrasp [9]. Contact-GraspNet predicts precise 6-DoF grasps from point clouds and performs well on complex geometries, while AnyGrasp directly outputs grasp configurations from depth images and is efficient in cluttered scenes. The system selects between these methods based on object geometry and scene complexity.

For motion planning, we use NVIDIA Isaac CuMotion integrated with MoveIt to generate collision-free trajectories in real time with GPU acceleration. The workspace of the 7-DoF xArm7 is constrained using URDF models of the robot and an OctoMap constructed from a grounded point cloud, either from the head-mounted Orbbec Femto Bolt RGBD camera or the realsense d435 anchored on the hand, reducing self-collision risks. This perception-to-grasping pipeline is deployed in tasks such as Storing Groceries and has demonstrated reliable performance in competition settings.

4 Domestic Task

This section provides a concise overview of how the above components are instantiated to solve specific RoboCup@Home tasks. We focus on task-level integration rather than repeating the detailed methods presented in the previous section.

4.1 Receptionist

Speech We use OpenAI Whisper [10] for offline automatic speech recognition and dynamic query of LLM API’s for more accurate understanding of human intent. In the Receptionist task, this combination allows Tinker to robustly extract guest names, interests, and beverage preferences from diverse phrasings, rather than relying on fixed command templates.

Face Recognition To support multi-user interaction, the robot maintains a simple face recognition pipeline with separate enrollment and recognition stages. During enrollment, the user stands in front of the camera while multiple facial embeddings are collected. During operation, a CNN-based recognizer built on ArcFace computes similarity scores between live facial embeddings and stored profiles, enabling Tinker to identify guests and match them to previously recorded names and preferences.

4.2 Storing Groceries

Manipulation In the Storing Groceries task, Tinker must pick up items from a table and place them in appropriate storage locations. Manipulation is handled by the 7-DoF xArm7, with motion planning executed through MoveIt and Isaac CuMotion as the planner plugin. The arm workspace is constrained using URDF-based collision models, and an OctoMap generated from the end-effector-mounted RealSense depth camera is used to avoid collisions with the environment and the robot base. Grasp candidates are generated using AnyGrasp [9] and filtered for reachability and stability before execution.

Plane Extraction To distinguish supporting surfaces such as tabletops from surrounding structures, we employ a real-time plane extraction module on organized RGB-D point clouds. By grouping pixels into local neighborhoods and clustering them into planar regions, the algorithm reliably identifies major planes in the scene at over 15 Hz for 1096×720 depth images. The resulting plane masks are used to segment objects resting on the table for subsequent grasping.

Object Recognition For object recognition within the Storing Groceries task, we build on the vision training pipeline described in Section 3.4. In case of previously unseen objects, GroundingDINO [6], with task-specific text prompts, is used to generate bounding boxes for target objects, and a lightweight SAM

variant refines these into segmentation masks. The corresponding 3D points from the depth image are then used to estimate object pose and size, enabling precise grasp and placement planning for the detected items.

4.3 Help Me Carry

Human Tracking The Help Me Carry task reuses the multimodal human tracking framework described in Section 3.2. The system maintains a stable identity lock on the designated operator and provides a 3D position estimate of the operator in the robot’s coordinate frame.

Follow Human following is implemented by treating the tracked operator pose as a moving goal within a SLAM-aware navigation framework. The robot continuously builds a local map, estimates its own pose, and projects the operator’s position into the map coordinate frame. Temporal prediction allows short-term motion continuity when the operator temporarily leaves the field of view, while a re-acquisition mechanism restores tracking when the operator reappears. By tightly coupling vision-based tracking with obstacle-aware navigation, Tinker achieves reliable Help Me Carry behavior in partially unknown indoor and semi-outdoor environments with minimal human intervention.

4.4 GPSR

LLM-Based Task Planning For the GPSR task, Tinker employs an LLM-based high-level planner that interprets unconstrained natural-language commands and produces executable robot behaviors in a closed-loop manner. The LLM takes as input the user command, a symbolic task state, a predefined action set, and prior knowledge, and outputs the next atomic action in a structured JSON format. Supported actions cover navigation (**goto**), perception (**scanfor**), manipulation (**grasp**, **place**), and human-robot interaction (**announce**, **listen**).

A symbolic knowledge base of semantic locations, object categories, and grasp primitives constrains and grounds the LLM outputs for safe execution. After each action is executed by the low-level controllers, the symbolic state is updated using sensor feedback and fed back to the LLM. This loop enables robust execution of long-horizon, multi-step tasks such as guiding a person, delivering objects across rooms, and answering queries in dynamic domestic environments.

```
COMMAND: go to the kitchen and ask James what the time it is
ACTION: 1. announce 2. goto 3. grasp 4. place 5. scanfor 6. listen
STATE: arrived at human position
According to the information given, return a JSON response following the format
below
- action: string
- parameters: object
for example: {{action: goto, parameters: {{target: bedroom}}}}
make sure to return a valid JSON and valid input parameter
```

((a)) LLM Prompt

```
{
  "action": "announce",
  "parameters": {
    "message": "Hello James, what time is it?"
  }
}
```

((b)) LLM Returned JSON

Fig. 2: GPT Spatial Reasoning

References

1. L. Ma, J. Meng, S. Liu, W. Chen, J. Xu, and R. Chen, “Sim2real2: Actively building explicit physics model for precise articulated object manipulation,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 11 698–11 704.
2. P. Xie, R. Chen, S. Chen, Y. Qin, F. Xiang, T. Sun, J. Xu, G. Wang, and H. Su, “Part-guided 3d rl for sim2real articulated object manipulation,” *IEEE Robotics and Automation Letters*, 2023.
3. S. Tao, F. Xiang, A. Shukla, Y. Qin, X. Hinrichsen, X. Yuan, C. Bao, X. Lin, Y. Liu, T. kai Chan, Y. Gao, X. Li, T. Mu, N. Xiao, A. Gurha, V. N. Rajesh, Y. W. Choi, Y.-R. Chen, Z. Huang, R. Calandra, R. Chen, S. Luo, and H. Su, “Maniskill3: Gpu parallelized robotics simulation and rendering for generalizable embodied ai,” 2025. [Online]. Available: <https://arxiv.org/abs/2410.00425>
4. M. Labbé and F. Michaud, “Rtab-map as an open-source lidar and visual simultaneous localization and mapping library for large-scale and long-term online operation,” *Journal of Field Robotics*, vol. 36, no. 2, p. 416–446, Oct. 2018. [Online]. Available: <http://dx.doi.org/10.1002/rob.21831>
5. Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, “Bytetrack: Multi-object tracking by associating every detection box,” 2022. [Online]. Available: <https://arxiv.org/abs/2110.06864>
6. S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, C. Li, J. Yang, H. Su, J. Zhu *et al.*, “Grounding dino: Marrying dino with grounded pre-training for open-set object detection,” *arXiv preprint arXiv:2303.05499*, 2023.
7. A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, “Segment anything,” 2023. [Online]. Available: <https://arxiv.org/abs/2304.02643>
8. N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, J. Lei, T. Ma, B. Guo, A. Kalla, M. Marks, J. Greer, M. Wang, P. Sun, R. Rädle, T. Afouras, E. Mavroudi, K. Xu, T.-H. Wu, Y. Zhou, L. Momeni, R. Hazra, S. Ding, S. Vaze, F. Porcher, F. Li, S. Li, A. Kamath, H. K. Cheng, P. Dollár, N. Ravi, K. Saenko, P. Zhang, and C. Feichtenhofer, “Sam 3: Segment anything with concepts,” 2025. [Online]. Available: <https://arxiv.org/abs/2511.16719>
9. H.-S. Fang, C. Wang, H. Fang, M. Gou, J. Liu, H. Yan, W. Liu, Y. Xie, and C. Lu, “Anygrasp: Robust and efficient grasp perception in spatial and temporal domains,” 2023. [Online]. Available: <https://arxiv.org/abs/2212.08333>
10. A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” 2022. [Online]. Available: <https://arxiv.org/abs/2212.04356>

Robot Tinker Hardware and Software Description

Mechanical specifications of Robot Tinker are as follows:

- **Base:** AgileX Tracer Mini, 2m/s max speed.
- **Torso:** Aluminum extrusions.
- **Arm:** Mounted on Base. UFACTORY Xarm7 for accessing objects. Maximum load: 5kg.
- **End-Effector:** Mounted on arm. UFACTORY gripper.
- **Head:** 3D-printed mount
- **Robot dimensions:** Width: 0.40m, Length 0.40m, Height: 1.6m (max).
- **Robot weight:** 50kg.

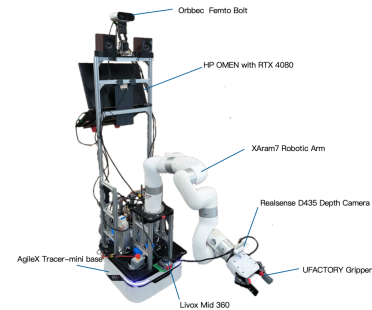


Fig. 3: Tinker

Also our robot incorporates the following devices:

- Femto Bolt - ORBBEC depth camera
- Realsense D435I camera
- MAONO PM461 Microphone
- Livox MID-360 laser scanner

Tinker Software Description

For our robot we are using the following software:

- **Platform:** Ubuntu 22.04 Operating System and ROS2 Humble.
- **Navigation:** ROS2 NAV2
- **Face recognition:** Acrface
- **Object recognition and segmentation:** GroundingDINO, SAM, YOLO-seg
- **Human tracking:** YOLO-track, custom ResNet*
- **Speech recognition:** Openai Whisper
- **Speech generation:** CoquiTTS
- **Manipulation:** MoveIt2, AnyGrasp, Cumotion
- **Simulation:** SAPIEN
- **LLM:** OpenChat, GPT-4o

External Devices

- **HP OMEN** with RTX 4080, i9-13900HX core